

BRAINERD LAKES, MN, USAWMO: **726555**Lat: **46.405N**Lon: **94.131W**Elev: **1221**StdP: **14.06**Time Zone: **-6.00 (NAC)**Period: **95-19**WBAN: **94938****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-13.3	-25.6	1.4	-17.5	-20.7	1.8	-11.3	22.5	15.2	20.3	15.1	3.3	240	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	88.1	71.9	84.6	69.2	82.1	68.1	74.9	84.1	72.7	81.5	70.6	79.0	11.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.1	124.1	81.1	69.8	114.8	78.5	67.4	105.6	76.4	39.2	84.2	37.1	81.2	35.2	78.9	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	18.5	16.7	-28.2	93.1	7.5	3.1	-33.6	95.4	-38.0	97.3	-42.2	99.0	-47.7	101.3		
(5)			-26.8	78.1	4.9	1.8	-30.3	79.4	-33.2	80.5	-35.9	81.5	-39.5	82.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	11.1	15.6	28.3	42.8	55.2	64.9	70.2	67.2	59.4	45.9	30.9	16.9
	DBStd	22.83	13.93	13.01	12.85	9.96	8.46	6.72	5.87	5.64	8.28	8.86	10.88	12.38
	HDD50	4950	1206	962	679	256	42	1	0	0	14	188	575	1027
	HDD65	8639	1671	1382	1139	668	325	82	19	39	206	594	1022	1492
	CDD50	2216	0	0	6	40	202	447	626	534	297	60	4	0
	CDD65	428	0	0	0	1	19	78	180	108	40	2	0	0
	CDH74	4036	0	0	1	21	269	758	1670	947	343	27	0	0
	CDH80	1112	0	0	0	3	77	186	518	244	81	3	0	0
(14) Wind	WSAvg	7.3	7.2	7.5	7.5	8.3	7.9	6.9	6.4	6.0	7.1	7.9	7.7	7.3
(15) Precipitation	PrecAvg	27.2	0.7	0.6	1.4	2.3	3.6	4.3	3.9	3.3	2.6	2.6	1.3	0.7
	PrecMax	38.2	1.6	1.4	3.5	5.3	8.7	7.3	9.7	7.3	6.2	6.3	4.1	1.8
	PrecMin	17.6	0.2	0.1	0.4	0.6	1.3	1.0	1.3	0.9	0.8	0.1	0.2	0.2
	PrecStd	4.6	0.4	0.3	0.7	1.0	1.7	1.7	2.0	1.7	1.3	1.7	1.0	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	47.2	66.3	76.6	87.9	89.8	92.9	90.6	86.7	78.2	63.3	44.7
		MCWB	36.9	41.1	55.1	57.5	67.0	72.6	74.8	73.3	69.9	60.8	52.5	39.3
	2%	DB	36.7	42.1	57.4	70.4	81.1	84.8	88.5	85.6	82.1	71.0	56.6	38.8
		MCWB	33.1	36.5	47.8	53.1	62.5	68.7	73.0	70.3	68.2	59.0	48.3	35.1
	5%	DB	33.7	37.1	50.6	65.5	75.5	81.6	85.1	82.2	78.3	64.5	51.4	35.8
		MCWB	31.7	33.6	43.6	49.7	59.3	66.6	70.3	68.2	65.7	54.7	44.4	33.4
	10%	DB	30.3	34.2	45.2	60.8	71.6	78.7	82.3	79.5	73.4	60.7	46.2	33.5
		MCWB	28.3	31.6	39.3	47.5	57.5	64.5	69.3	66.8	63.8	51.9	41.1	31.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	41.6	58.2	60.0	70.3	75.8	78.1	76.7	73.4	65.5	55.3	40.0
		MCDB	43.1	46.3	64.4	73.2	83.1	85.9	87.5	86.4	81.9	72.4	59.4	42.2
	2%	WB	33.8	37.2	49.2	55.0	65.6	72.1	75.6	73.3	70.7	60.2	48.9	35.6
		MCDB	36.0	41.2	55.8	66.4	76.3	81.5	84.9	81.9	78.9	68.3	55.5	38.0
	5%	WB	31.4	34.2	44.1	51.4	62.2	69.3	73.5	70.9	68.1	56.4	45.0	33.4
		MCDB	33.9	37.2	50.8	63.7	72.3	77.7	82.6	79.3	74.8	63.9	50.3	35.3
	10%	WB	28.4	31.4	39.4	48.3	59.0	66.7	71.3	68.7	65.6	53.0	41.4	31.3
		MCDB	30.1	34.1	44.6	59.1	68.3	75.1	79.5	76.4	72.7	59.0	46.4	33.2
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	19.3	19.3	20.8	21.7	20.4	20.6	20.6	20.1	17.5	15.0	14.8
		MCDBR	17.7	19.2	24.5	30.4	29.4	24.8	23.4	24.1	25.2	25.1	22.9	16.1
		MCWBR	14.6	14.9	15.7	16.6	14.4	11.9	10.9	11.5	13.0	14.4	15.2	13.7
	5% WB	MCDBR	17.2	17.6	23.6	26.6	24.4	20.4	21.1	20.6	20.2	21.6	20.6	14.6
		MCWBR	14.3	13.8	15.5	15.9	14.0	11.5	11.6	11.3	12.1	13.9	14.6	12.8
(40) Clear-Sky Solar Irradiance	taub		0.256	0.282	0.308	0.360	0.384	0.379	0.379	0.388	0.357	0.311	0.284	0.261
	taud		2.382	2.312	2.384	2.331	2.323	2.366	2.386	2.376	2.437	2.558	2.514	2.411
	Ebn at Noon		272	284	292	284	280	281	279	272	272	273	260	257
	Edh at Noon		22	30	32	37	39	38	36	36	30	23	20	19
(44) All-Sky Solar Radiation	RadAvg		510	864	1213	1512	1700	1862	1979	1660	1249	771	479	355
	RadStd		58	86	118	133	127	125	81	137	93	112	48	41

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	-2.48	-2.23	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page